Module 3 — Read 0

Project preview — descriptive analysis toolbox

Module 3, Read 0: Describing Patterns

Big question: What does a dataset *look like*, and what patterns are inside it?

Project: Project 3 — Discovering Patterns · Next: Read 1 · R patterns: R reference — Module 3

> Short read — start here
>
> Five-minute orientation before Read 1. Modules 1 and 2 focused on inference and data collection for one variable at a time. Module 3 is a complete introduction to descriptive analysis: organizing graphs and numeric summaries by how many variables you have and what type each one is. A descriptive analysis does not try to generalize to a larger population with the tools of hypothesis tests or confidence intervals from the previous modules.

Good science: describe before you generalize

In Module 0, Read 0 you met the five-step statistical process. Modules 1–2 used it for testing and estimating one parameter. Project 3 uses the same steps to explore and describe a multivariable dataset with graphs and summaries, not claims or estimates for a larger population.

Step	In Project 3
1 — Ask	What patterns might live in this dataset? Which variables and pairs are worth describing?
2 — Plan	Pick a documented source (e.g. TidyTuesday); choose four variables; classify each as numerical or categorical; decide which graph/summary fits each variable-type combination (milestone).
3 — Collect	Download or load the dataset; record source URL; check row count and missing values.
4 — Analyze	In your notebook: one graph + numeric summary per variable, and the right bivariate graph/table for all six pairs. Use R (`tidyverse`, `ggplot2`) — patterns taught in Demo 1 and Demo 2.
5 — Conclude	One-page report: two best graphics, 4 labeled numeric facts, four findings in plain language from descriptive tools only.

Statistics moves among three worlds (Module 0). In Project 3 you stay mostly in observe and compute: you are not required to infer population parameters.

1. The world we imagine (questions first)

- Name variables you care about and their types (numerical vs categorical).
- Choose graphs from the descriptive toolbox (bar, histogram, boxplot, scatter, stacked bar) before you open Colab and start writing R code.

2. The world we observe (data second)

- Load the real dataset; inspect with `glimpse()`.
- Compute counts, proportions, means/medians, IQR/SD from your rows.
- Every statistic is a description ($\hat{p}$, $\bar{x}$, median, r) of our observed world.

3. The world of computing (R connects plan to data)

- `count()` / `summarize()` — numeric summaries by group.
- `ggplot2` — `geom_bar`, `geom_histogram`, `geom_boxplot`, `geom_point`, stacked bars.
- `cor()` — Pearson r when a scatterplot looks roughly linear (introduced in Read 2).

The story

In Modules 1–2 you asked inferential questions: *Is* p *different from* p_0? *What is a plausible value for* p *or* μ? In Module 3 you will explore a real dataset with several variables by making graphs and computing numeric summaries for each variable and then for pairs of variables together.

Exploratory data analysis (EDA) builds a picture of what the data contain in your file. Project 3 is descriptive only but in the readings I'll introduce a few more inferential tests (t-test, chi-square, etc.) that could be used to answer questions like we did in Project 1 and 2 that are about claims or estimates of a population.

Variable types

As we start to have many tools available for analyzing data, it will be helpful to keep them organized. One important characterization for this purpose is the **type** of variable, that is, when we ask a question and obtain answers from each item in our sample, what *type* of answer did we get? A number or a word? The official terms for these two basic types are **numerical** and **categorical**.

Type	Definition	Examples
Categorical	Values are labels for each category; distance between labels is not meaningful for arithmetic	Species, island, sex, yes/no
Numerical	Counts or measurements where order and distance matter	Flipper length (mm), body mass, age

Binary categorical = exactly two levels/categories (yes/no, male/female). Ordinal categorical = levels/categories can be ordered (but distance between them is not meaningful) like happy/sort of happy/sad/very sad.

How many variables? Descriptive toolbox

Organize your descriptive tools by number of variables and type:

# Variables	Variable types	Numeric summary (statistic)	Graph
1	Categorical (incl. binary)	Count x, proportion $\hat{p} = x/n$ (or %)	Bar chart or pie chart
1	Numerical	Center ($\bar{x}$, median) + spread (SD, IQR, range)	Histogram, dot plot, boxplot
2	Numerical × categorical	Group-wise center & spread (`group_by` + `summarize`)	Side-by-side boxplots, grouped histograms
2	Categorical × categorical	Two-way counts; joint / row / column proportions	Stacked bar (counts or 100% fill)
2	Numerical × numerical	Form, direction, strength; Pearson r when linear	Scatterplot

See also the Describe rows in the methods map.

Reading path (Module 3)

Step	Reading	Topics
Read 1	Part 1	§1 One categorical · §2 One numerical · §3 Numerical × categorical
Quiz A	Def & Notation	After Read 1 · by hand + graph interpretation
Read 2	Part 2	§4 Two categorical · §5 Two numerical
Quiz B	Concept Check	After Read 2 · scatter, two-way tables, Pearson r
Labs	Demo 1 → Demo 2 → Exercise	Tidyverse in R
Synthesis	Integration quiz	After exercise lab

Tour the sample project

Open the sample Project 3 — Palmer penguins (example page). It mirrors the structure of your one-page report using a TidyTuesday-style dataset.

Section	What to notice	World
Introduction	Dataset source, four variables named with types	Imagine
Univariate	One graph + one numeric fact per variable	Observe
Bivariate	Right graph/table for each pair; plain-language findings	Observe
Notebook	Runnable R from load → graphs → summaries	Compute

Requirements for Project 3 dataset

- 4 variables of type numerical or categorial (no image or long text data please)
- 10 rows of observational units (more is fine)
- A documented source (TidyTuesday, government open data, your own data, a friend's research study, etc.)
- Variables with a real-world story you're interested in spending several hours thinking about

What to do next

1. Glance at Project 3, the Phase A milestone, and the sample example; note due dates on the course path.
2. Browse a few TidyTuesday datasets — pick one that interests you.
3. Start Read 1 — Section 1 (one categorical), then Section 2 (one numerical).

When you finish Read 1, take Quiz A, then continue with Read 2.

Check yourself — Module 3 preview

1. In one sentence: how is descriptive analysis different from the inferential work in Modules 1–2?
2. Name the five steps of the statistical process in order.
3. For one categorical variable, name one graph and one numeric summary.
4. For two numerical variables, name the first graph and three words you use to describe a scatterplot.
5. Where will you learn the R code for these graphs — reads or demo labs?

Answers — Module 3 preview

1. Descriptive analysis summarizes your dataset as observed. Inferential analysis tests or estimates population parameters.
2. Ask → Plan → Collect → Analyze → Conclude.
3. Bar chart; counts or proportions.
4. Scatterplot; form, direction, strength.
5. Demo labs teach code; reads show interpretive output.